DESIGN DETAIL DOCUMENT_TEAM_GECCS

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Introduction:

The following document aims to elaborate on the points mentioned in the Solution Ideas form (for DD Robocon 2024). It enlightens the design aspect of both the robots concerning to CAD models and Simulation with a logical approach and covers each point concisely. The prime objective of this design is to fabricate lightweight, sustainable and fail-proof robots under the mentioned constraints and complete the assigned task with maximum efficiency in minimum time.

I. DESIGN OF ROBOT-1(R-1):

Robot-1 (R-1) is semi-autonomous with holonomic 4-wheeled drive consisting Omni wheels, controlled by a Bluetooth 5.0-enabled wireless controller for doing tasks in Area 1 and Area 2 in given time constraints. It features four pneumatic grippers for precise Seedling picking and planting. Three rollers gather Paddy Rice and Empty Grain, feeding them to a Transferring mechanism with two counter-rotating wheels, ensuring seamless transfer to the Storage Zone in Area 3.

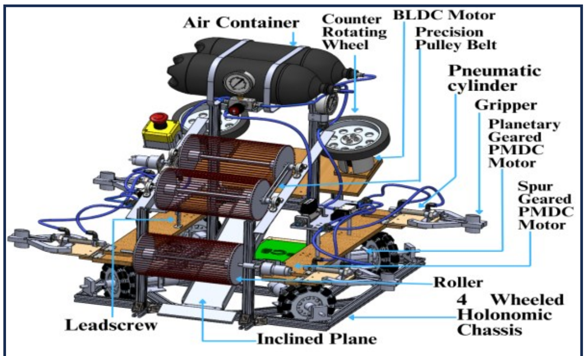

Fig.1. *Isometric view of Robot-1*

A. Overall Dimensions (in mm) and Estimated weight (in kg's)

Table 1: *Dimensions of R-1*

Parameter	Value	Extended
Length	698.50mm	698.50mm
Width	580.00mm	894.02mm
Height	691.66mm	691.66mm
Estimated Weight	24Kg	

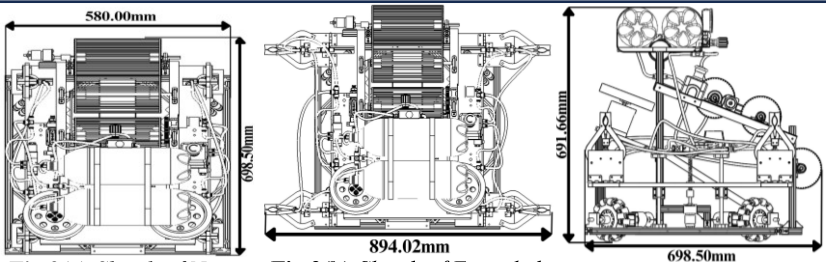

Fig.2(a) *Sketch of Non-extended Top view* **Fig.2(b)** *Sketch of Extended Top view* **Fig.2(c)** *Sketch of Side view*

B. Type of Drive:

- R-1 is a semi-autonomous robot utilizes a 4-Wheeled drive for holonomic kineticism, enhancing its manoeuvrability and efficiency for high-speed performance of wheels.
- All four omni wheels, with a 127mm diameter and fixed orientation, provide instantaneous tracing in any direction. Mounted at a 45° angle on the chassis, their velocity components contribute to motion along x and y directions.
- The relationship between the wheels' speed v_1, v_2, v_3, v_4 and the robot's speed v_x, v_y and ω is described by the equation:

$$\frac{1}{\sqrt{2}}\begin{bmatrix}-1 & -1 & 1 & 1\\ 1 & 1 & -1 & 1\end{bmatrix}\begin{bmatrix}v_1\\ v_2\\ v_3\\ v_4\end{bmatrix}=\begin{bmatrix}v_x + & d\cdot\omega\\ v_y + & d\cdot\omega\end{bmatrix}$$

(d is the distance between wheels and R-1's Chassis centre)

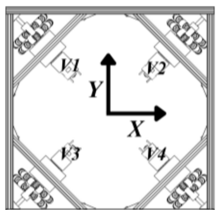

Fig.3. *Omni Wheeled Chassis*

- The R-1's stable setup is ensured by the center of mass (COM) at X=222 mm, Y=409 mm, Z=555 mm from origin as per CAD model, promoting balance and preventing tipping or instability.

C. Type of Actuators integrated:

ACTUATORS	APPLICATIONS	SPECIFICATIONS	QTY
Planetary Geared PMDC Motor	Locomotion of Chassis	Base Motor RPM: 12600, Load Torque: 1.47N-m, Stall torque: 13.5N-m, Stall current: 59A, Reduction ratio=16:1	4
BLDC Motor	To drive the counter-rotating wheels (Transferring Mechanism)	KV Rating: 200KV, Power: 1200W, Stall current: 46A, Voltage Range: 24V-48V	2
Spur Geared PMDC Motor	To drive rollers and leadscrew (Picking and feeding mechanism)	Base Motor RPM: 18000, Operating Voltage: 6-18 V, Stall Torque: 1.22 N-m, Reduction Ratio=30:1	4
Pneumatic Cylinder	Picking Seedlings (Gripper mechanism)	Bore Diameter: 12mm, Stroke Length: 25mm, Operating Pressure 0.05 to 1Mpa	4

D. Type of Sensors integrated:

SENSORS	APPLICATIONS	SPECIFICATIONS	QTY
9 Axis Absolute Motion Shield	Maintain orientation of Chassis	Operating Voltage: 5V, Power consumption: 50mW, Combination of Accelerometer, Gyroscope and Magnetometer	1

E. Seedling picking and planting mechanism:

- **Objective:** To pick up Seedlings from the Seedling Rack and plant them in the Planting Zone.
- **Process Flow (Mechanism):**

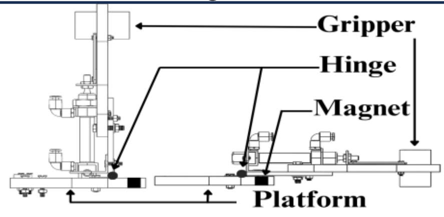

Fig.4(a) *Folded gripper* **Fig.4(b)** *Extended gripper*

- The Seedling picking mechanism comprises of two platforms on either side of R-1, each with two pneumatic grippers for efficient Seedling handling *[fig.6]*.
- Grippers are manually folded before gameplay*[fig.4(a)]* and extend after the game begins due to the inertia of the robot*[fig.4(b)]*. Then the gripper sticks to the magnet which is mounted on the platform. Both the gripper and platform have 2 degrees of freedom.

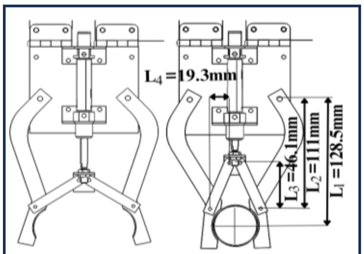

Fig.5(a) *Actuated State* **Fig.5(b)** *Retracted State*

- As the piston actuates, it pushes the jaws apart , which results in the opening of the gripper due to linkages *[fig.5(a)],* and as the piston retracts, it grips seedlings from the Seedling Rack *[fig.5(b) & fig.8(a)]*. The platform moves upward using a leadscrew and motor to lift the Seedling from the Seedling rack*[fig.8(b) & fig.6*]
- Both grippers on the platform retract simultaneously, collecting two Seedlings at a time. R-1 changes orientation by 180^0, gathers the next two Seedlings by second Platform. After picking four Seedlings, R-1 moves to the Planting Zone [fig.8(c)]. Moving the platform downward and actuating the piston [fig.8(d)] results in planting two seedlings per platform. The process repeats for the second platform.

- **Design approach (Justification):**

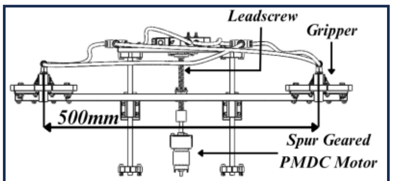

Fig.6. *Platform*

- The distance between two pneumatic grippers of a platform is 500mm *[fig.6]*. The gripper's encompassing shape increases stability and reduces force required to grip the seedling.
- The pneumatic cylinders are operated by two direction control valves actuated through a 24V DC supply, each controlling two grippers on the same platform *[Fig.7]*.
- The Pneumatic Cylinder's pressure is set at Operating pressure (P)= 0.3Mpa, which is regulated by using a pressure regulator from the Air Container of 0.6Mpa. Bore diameter = 12mm, bore area(A)=113.1mm^2.
  ∴The piston force is **F = P x A= 33.93N**

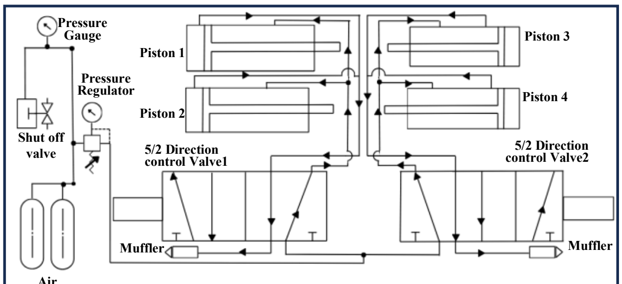

Fig.7. *Pneumatic Schematic Diagram*

- **Force analysis of gripper mechanism:**
  Following data relate to a mechanical gripper using friction to grasp an object: $L_1 = 128.5$mm, $L_2 = 111$mm, L_3 =46.1mm, L_4 =19.3mm (where L_1, L_2, L_3, L_4 are length of linkages of gripper)*[fig.5(b)],*Multiplying factor (K)=0.361, Factor of Safety (Z) =1.4, No. of fingers(n_f) = 2, Co-efficient of friction of PVC to rubber (μ)=0.18, Weight of Seedling (W)=mg=3.28N (Mass of Seedling (m)=0.335kg)

$$\therefore \textbf{Gripping Force } (F_g) = \frac{zw}{\mu n_f}(1+k) = \textbf{17.35N} \quad \therefore \textbf{Actuating Force } (F_a) = 2F_g \times \left[\frac{L_1}{L_2} \times \frac{L_3}{L_4}\right] = \textbf{95.76 N}$$

$$\therefore \textbf{Required force by piston } (P_r) = F_g\left[\frac{L_1}{L_2} \times \frac{L_3}{L_4}\right] = \textbf{47.88 N}$$

- The gripper has a gripping force of 17.35N, surpassing the required force 3.28N to lift the seedling. Therefore, the gripping force is sufficient for securely holding the seedling.
- The total weight of the individual platform, including two grippers and two seedlings is 2.17 kg. It is lifted by a leadscrew and a spur-geared PMDC motor with a stall torque of 1.22 N-m, allowing to effortlessly lift the platform with seedlings and move vertically upward. The trapezoidal 4-start leadscrew has a diameter of 8mm, a pitch of 2mm. Platform travel 94mm for up and down movement, taking 1.5 sec.
  ∴ Thus, a speed of 0.062 m/s is achieved to move the platform up and down.

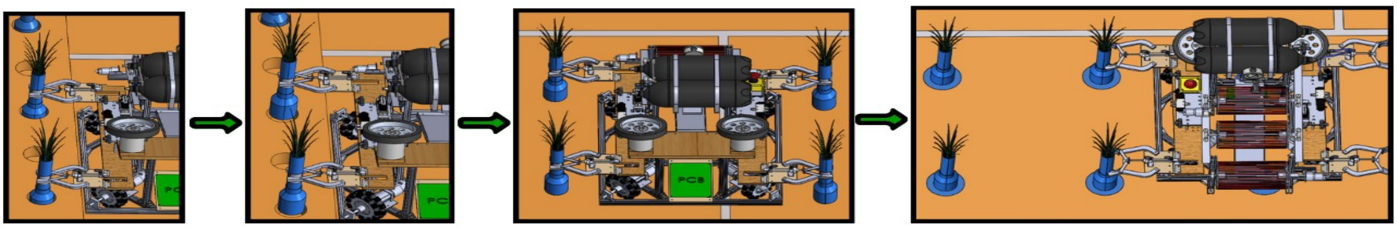

Fig.8(a) *Gripped Seedlings* **Fig.8(b)** *Picked Seedlings* **Fig.8(c)** *Moving towards Planting Zone* **Fig.8(d)** *Planting two Seedling at a time*

F. Paddy Rice, Empty Grain picking and transferring mechanism:

- **Objective**: Pick the ball (Paddy rice and Empty grain) from the Harvesting Zone and transfer it to the Storage Zone of Area 3.
- **Process Flow(mechanism):**

- Ball Picking Mechanism comprises three rollers: Bottom, Middle, and Top Roller.
- Thus, Bottom Roller picks the ball from the Harvesting Zone [*fig.9(a)*], Middle and Top Roller stores ball until R-1 reaches to water zone and then feed the ball to the transferring mechanism. [*fig.9(b)*].
- Then transferring mechanism safely throw the ball to storage zone of Area 3 [*fig.9(c)*], due to wheel speed, frictional force, and stored elastic potential energy [*fig.9(d)*].

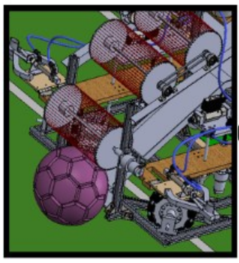

Fig.9(a) *Picking the ball*

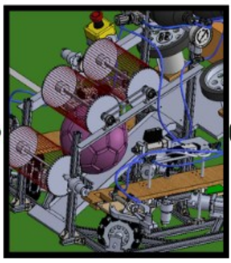

Fig.9(b) *Ball feeding to transferring mechanism*

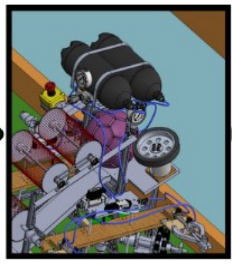

Fig.9(c) *Compressed ball between two counter rotating wheels*

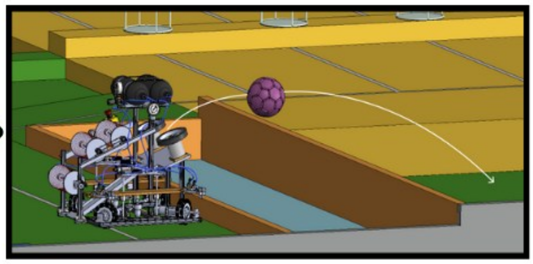

Fig.9(d) *After releasing the ball to Storage Zone (ball trajectory)*

- **Design approach of ball picking mechanism (Calculations):**

- The roller comprises of two parallelly separated circular discs and are connected using a D-shaft at the centre of the discs (3D Printed Disc). Several elastic rubbers are stretched between the discs along their circumference, forming a cage-like structure.
- All three rollers are above the inclined plane which are driven by two Spur Geared PMDC motors: Motor1 drives Bottom Roller, Motor2 drives Middle and Top Roller. An inclined plane facilitates collection of ball and further guides the upward movement of the ball, ultimately reaching to the counter-rotating wheels.
- Motor2 is mechanically coupled to the top roller, and motion is transmitted to both the top and middle rollers through a precision pulley belt drive system.

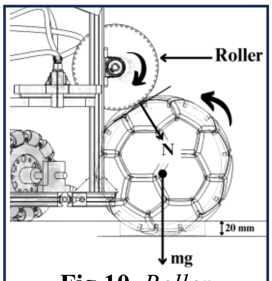

Fig.10. *Roller*

- Transferring mechanism comprises two counter-rotating wheels driven by two BLDC motors.
- In order to determine the required parameters for the design of roller, following calculation are required:

a) **Gripping force(T)** acting on the ball due to elastic rubber. Here A (Area of rubber) $=1.96\times10^{-5}\text{m}^2$, E (Young's modulus of rubber) $=1.27\times10^{6}\text{N/m}^2$, Lo (Original length of rubber) =0.075m, L_1(length of rubber when stretched between two discs) =0.215m, L_2(Length of rubber under tension of ball) =0.250m, ΔL(Change in length of rubber) =0.175m, $\mathbf{T=A\times E\times \frac{\Delta L}{L_o}=58.08N}$

b) **Linear velocity of roller (V)** Here R (r.p.m. of motor) =410rpm, r (radius of roller) =0.06, **ω** (Angular velocity of roller) $\boldsymbol{\omega=\frac{2\pi R}{60}=42.94\ rad/s,\ V=\omega r=2.576m/s}$

c) **Frictional force** ($\boldsymbol{Fr}$) between ball and rubber of roller, For μ = 1.15, $\boldsymbol{F_r=\mu T=66.792\ N}$

d) **Centrifugal forces** due to roller on ball, mass of roller (M)=0.300Kg, $\boldsymbol{F_c=\frac{Mv^2}{r}=33.18N}$

- **Design approach of ball transferring mechanism (Calculations):**

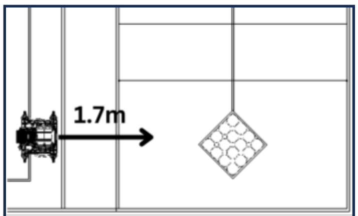

Fig.11. *Range of trajectory*

- To calculate the launching velocity of the ball, the following trajectory equation is employed, $\mathbf{y}=\boldsymbol{x\tan\theta-\frac{gx^2}{2u^2\cos^2\theta}}$
- Once the ball is collected, R-1 transports the ball to the Storage Zone, considering a max range of trajectory **(x) of 1.7m** from R-1 and the ball's approaching height **(y) of 0.42517m** at a launching angle **($\boldsymbol{\theta}$) of 22°**. Substituting the above values into the equation of the trajectory, **u=7.87m/s**. The mechanism of counter-rotating wheels is employed to achieve the required velocity and accuracy in the throw.
- To successfully cross the wall of water zone, the ball must overcome a height barrier of 0.2 meters, to achieve max height of trajectory (H). Where g is acceleration due to gravity 9.8m/s^2, $\text{H}=\boldsymbol{\frac{u^2\sin\theta^2}{2g}}=\mathbf{0.443m}$
- The main forces acting on the ball are discussed as follows:

a) **The normal force (*N*)** due to the compression of the ball while being released through the counter rotating wheels: Here E (Young's Modulus of Rubber) $=100\times10^{6}\ \text{N/m}^2$, s(surface area in contact)$=5.7\times10^{-5}\text{m}^2$

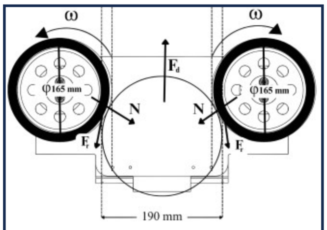

Fig.12. *Forces acting on ball*

,Δ_1(compression on ball) = 0.03m and r (radius of ball) = 0.095 m. Substituting the given values in the equation below, $N = \frac{E_s \Delta_1}{2r}$ = **900N**

b) Frictional Force (*Fr*) between the ball and the wheels. For μ = 0.67 (Outer surface material of wheels is Polyurethane), ***Fr* = 2μN =1206 N**

c) Centripetal force applied by the wheels on the ball. For the given values of m_c = 0.362kg, r_c = 0.0825m and u = 7.87m /s, $\mathbf{F_c = m_c \frac{u^2}{r_c}}$ **= 271.77N**

- To calculate the net **driving force (*F_d*)** of the system, work-energy theorem can be applied as follows, $\mathbf{I\omega^2 + (F_d - F_r)\, d = \frac{1}{2} M u^2}$
- Here, 'I' mass moment of inertia of wheel = 1.083×10^{-4} kgm^2, angular velocity(**ω**)= 75rad/sec, d is contact length of ball and wheels = 0.07 m, M is mass of the ball =0.350Kg. Substituting these values we get, ***Fd* = 1352.13N.**
- Now, to provide a driving force of **1352.13N,** a motor of **716.19 rpm** is required. As the **BLDC motors provide better controllability and variable characteristics**, they satisfy the need.

II. DESIGN OF ROBOT-2 (R-2):

Robot 2 (R-2) is fully Autonomous by operating line following and image processing in ROS frame work. It is designed for performing tasks in Area 3 in given time constraints. It is equipped with three rollers to gather Paddy Rice from Storage Zone, and it efficiently stores the collected Paddy Rice inside the Silo.

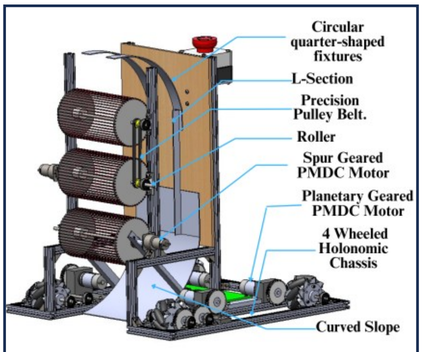

Fig.13. *Isometric View of Robot-2*

A. Overall Dimensions (in mm) and Estimated weight (in kg's)

Table 2: *Dimension of R-2*

PARAMETER	VALUE
Length	600.00mm
Width	600.00mm
Height	695.00mm
Estimated Weight	16Kg

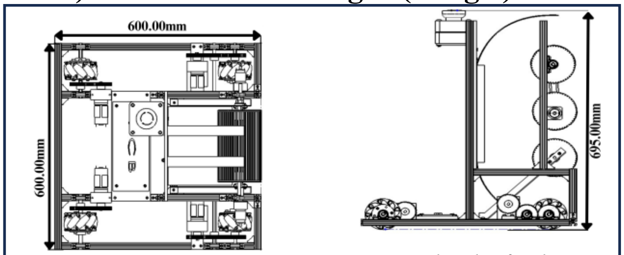

Fig.14(a) *Sketch of Top View* **Fig.14(b)** *Sketch of Side View*

B. Type of Drive

- R-2 is an autonomous robot that employs a 4-wheeled drive with holonomic kineticism, featuring mecanum wheels of 100mm diameter. These wheels contribute to stability and accuracy during straight-line motion, enhancing overall performance.
- The motors are coupled with the wheels through an external gear drive having gear ratio of 1:1, maintaining a balance between torque and speed.
- The relationship between the wheels' speed v_1, v_2, $v_3$and v_4, and the robot's speed v_x, v_y and ω is described by the equation: $\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \frac{1}{R} \begin{bmatrix} -1 & 1 & (a+b) \\ 1 & 1 & -(a+b) \\ -1 & 1 & -(a+b) \\ 1 & 1 & (a+b) \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ \omega \end{bmatrix}$

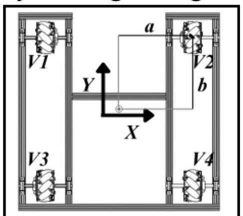

Fig. 15. *Mecanum Wheeled Chassis*

(Where, 'a' and 'b' represent half the Robot's width and length of Chassis from R-2's centre)

- The R-2's stable setup is ensured by the center of mass (COM) at X = -26mm, Y = 119mm, Z = -17mm from origin as per CAD model, promoting balance and preventing tipping or instability.

C. Type of Actuators integrated:

ACTUATORS	APPLICATIONS	SPECIFICATIONS	QTY
Planetary Geared PMDC Motors	Locomotion of Chassis.	Base Motor RPM: 9000, Stall torque: 5.26N-m, Reduction ratio=19.2:1	4
Spur Geared PMDC Motor	To drive rollers (Picking mechanism)	Base Motor RPM: 18000, Operating Voltage: 6-18 V, Stall Torque: 1.22 Nm, Reduction Ratio=30:1	2

D. Type of Sensors integrated:

SENSORS	APPLICATIONS	SPECIFICATIONS	QTY
LSA08	For line following	Operating Voltage: 7.5 to 20V, Sensing distance: 1 to 5 cm.	2
TF Mini LiDAR	To fetch the distance of ball from Robot-2	Measurement range: 0.1-12 ± 1% m, Operating Voltage: 5V, Power: 0.12W	1
Incremental Optical Rotary Encoder	For XY co-ordinates of Robot	Operating Voltage: 5V, Counts Per Revolution: 1600, 2 phase encoders	4
9 Axis Motion Shield	To maintain orientation of Chassis	Operating Voltage: 5V, Power consumption: 50mW, Combination of Accelerometer, Gyroscope and Magnetometer	1

Smart Vision Sensor-Object Tracking Camera	Detection of Paddy Rice and Empty Grain	Processor: NXP LPC4330, 204 MHz, dual core, Power consumption: 140 mA, Power input: USB input (5V) or unregulated input (6V to 10V), RAM: 264KB, Flash: 2MB	1

E. Seedling picking/planting and Paddy Rice, Empty Grain picking/transferring mechanism:

R-2 focuses on Paddy Rice detection, collection, and storage in silo of Area 3, while R-1 is skilled in seedling picking, planting and transfer Paddy Rice and Empty Grain to storage zone. They collaborate across Areas 1 and 2 to achieve "Mùa Vàng" by optimizing their performance through strategic division of responsibilities, ensuring effective and timely completion of tasks.

F. Collecting Paddy Rice and Storing in Silo mechanism:

➢ **Objective:** To collect the ball (Paddy Rice) from Storage Zone and store it into the Silo.

➢ **Process Flow (Mechanism):**

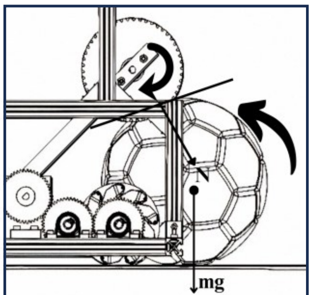

Fig.16. *Roller*

- The R-2 starts by following a line to reach Area 3. Then, it uses its onboard camera to spot Paddy Rice, employing smart algorithms to navigate and collect the Paddy Rice autonomously.
- R-2 uses odometry data from its encoders, comparing and filtering it with information from the Line Following Sensor (LSA08). This integrated approach ensures precise and refined outcomes.
- Collecting and Storing Paddy Rice mechanism comprises three vertically aligned rollers: Bottom Roller, Middle Roller, and Top Roller.
- The Bottom Roller collects Paddy Rice from Storage Zone *[fig.16 &fig.18(a)]*, and the Middle and Bottom Rollers hold it until R-2 reaches the Silo *[fig.18(b)]*. Once aligned with Silo, the Paddy Rice is stored safely inside Silo *[fig.18(c)& fig.18(d)]*.
- R-2 cyclically repeats the process, until achieving the "Mùa Vàng".

➢ **Design Approach (Calculation):**

- R-2 uses rollers similar to those in Robot- 1, with a curved slope at the base for easy ball slide-up. Two L-sections are mounted parallel to rollers to guide the balls in vertically upward motion. Motors are directly coupled to the roller to avoid power transmission losses. At the top, two Circular quarter-shaped fixtures guide the ball into the Silo for storage. The robot's design ensures proper ball storage and prevents them from falling out of silos or into the opposite team's area.
- Bottom Roller converts rotational motion into linear motion for Paddy Rice collection.
- The rubber roller applies a normal reaction force to the paddy rice, gripping it inward and propelling it further without any slippage.

In order to determine the required parameters for the design of roller following calculation are required:

a) Gripping force (T) acting on the ball due to elastic rubber. Here A (Area of rubber) $=1.77\times10^{-5}m^2$, E (Young's modulus of rubber) $=1.27\times10^{6} N/m^2$, Lo (Original length of rubber) =0.080m, L_1(Length of rubber when stretched between two discs) =0.220m, L_2(Length of rubber under tension of ball) =0.265m ΔL(Change in length of rubber) =0.185m *[fig.17]*, ∴ $\mathbf{T=A.E.\frac{\Delta L}{L_o}=50.57N}$

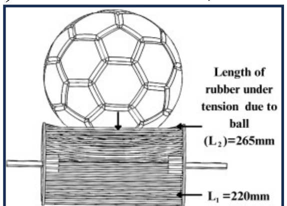

Fig.17. *Ball Picking*

b) Linear velocity of roller (V) Here R (r.p.m. of motor) =410rpm, r (radius of roller) =0.06, **ω** (Angular velocity of roller) $\boldsymbol{\omega=\frac{2\pi R}{60}=42.94}$ **rad/s, ∴ V= ω.r = 2.576m/s**

c) Frictional force (Fr) between ball and rubber of roller, For $\mu = 1.15$, $\boldsymbol{F_r = \mu T = 4.462}$ **N**

d) Centrifugal force due to roller on ball M (mass of roller) =0.300Kg, $\mathbf{Fc=\frac{Mv^2}{r}=33.18N}$

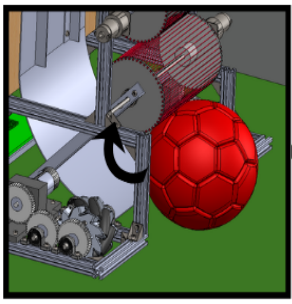
Fig.18(a) *Picking the Paddy Rice*

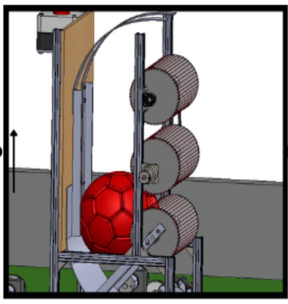
Fig.18(b) *Holding the Paddy Rice*

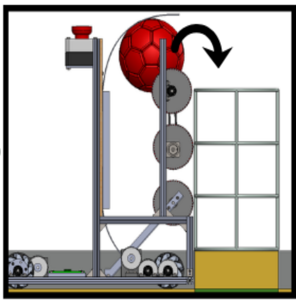
Fig.18(c) *Storing Paddy Rice into Silo*

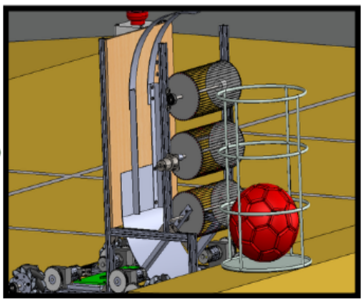
Fig.18(d) *Paddy Rice successfully stored in Silo*

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